INGESTION · DOMAIN 2 - DATA INGESTION AND ACQUISITION

Formats and sources

Section 2. Eight file formats, four message buses, and which mechanism reads them.

SECTION 2 of 10 WEIGHT 7% TOPIC Ingestion

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Some formats **carry** their schema. Others make you guess. Almost every ingestion decision follows from which kind you have.

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Eight file formats, in two groups

FORMAT	WHAT YOU GET
Parquet	Columnar, compressed, schema embedded. The default for analytics at rest.
ORC	Columnar, schema embedded. Common when arriving from a Hadoop estate.
Avro	Row-oriented, schema embedded. Common on message buses.
JSON	Text. Schema must be inferred or supplied. Nested by nature.
CSV	Text. Schema must be inferred or supplied. Flat and fragile.
XML	Text. Schema must be inferred or supplied. Deeply nested.
Text	One string column. You do the parsing.
Binary	The file's bytes, plus path and metadata. Nothing is parsed.

ON THE EXAM

The same eight are supported by Auto Loader and by COPY INTO. If a question offers a format outside this list, that is the discriminator.

What Parquet, ORC and Avro give you

- The schema is **read from the file**, not guessed. No sampling, no inference cost.
- Columnar formats let a query **read only the columns it needs**, which is why they suit analytics.
- Avro is row-oriented instead, which suits writing one record at a time rather than scanning columns.
- They are not human-readable, so debugging a bad file means tooling rather than a text editor.

> ON THE EXAM
>
> Row-oriented against columnar is the Avro versus Parquet question. Write one record at a time, or scan a few columns across many rows.

Where the cost and the risk live

- Schema must be **inferred from a sample** or supplied. Inference reads data before any work happens.
- A column that looks numeric in the sample and is not later will fail, or land in rescued data.
- Supplying a schema removes both the cost and the guesswork, at the price of maintaining it.
- JSON and XML are nested; CSV is flat. That shapes the transformation work far more than the parsing.

> ON THE EXAM
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> Schema evolution is only interesting for text formats. A Parquet file that gains a column tells you so; a CSV file just has an extra comma.

Text and binary as formats

- The **text** format returns one string column per line. Log files and anything with a custom layout.
- The **binary file** format returns the file's **content, path, length and modification time**.
- That is how you ingest images, PDFs and audio into a table without parsing them.
- Do not use them as a shortcut for a real format. That bypasses inference, evolution and rescued data.

> ON THE EXAM
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> Databricks' stated guidance is to set the real format rather than reading everything as text or binary first. Doing so pushes parsing into your own code.

Four ways to get files in

MECHANISM	CHARACTER
Auto Loader	Streaming source. Incremental, exactly-once, scales to billions of files.
COPY INTO	SQL command. Idempotent, re-triable, skips files already loaded.
read_files	SQL table-valued function. Reads files directly in a query.
Lakeflow Connect	Governed connectors. Extends Auto Loader to sources such as SFTP.

ON THE EXAM

Auto Loader and COPY INTO support the same eight formats. What differs is how they track what has already been read.

Auto Loader against COPY INTO

AUTO LOADER

A stream over a directory

- Tracks files in a RocksDB store in the checkpoint
- Handles millions of files per hour
- Schema inference, evolution and rescued data
- Recommended inside Lakeflow pipelines

COPY INTO

A command you run

- Tracks loaded files in the target table
- Idempotent: reruns skip what was loaded
- Schema inference, mapping, merging, evolution
- Best for a bounded set of files

THE CURRENT ADVICE

Databricks now recommends that SQL users reach for streaming tables rather than COPY INTO, for a more scalable and robust ingestion experience.

Four streaming sources

SOURCE	NOTES
Apache Kafka	The reference implementation. Event Hubs also connects through it.
Amazon Kinesis	AWS-native. Enhanced fan-out mode matters for low latency.
Google Pub/Sub	Read through a streaming function, inside pipelines.
Apache Pulsar	Supported through a streaming function, in Public Preview.

ON THE EXAM

A message bus is unbounded and append-only, so a streaming table is nearly always the right target for one.

Kafka gives you bytes, not data

- The key and value are **always returned as byte arrays**. Always, regardless of what was written.
- You must **cast to string**, or use a function such as from_avro, to get anything usable.
- The reader also returns topic, partition, offset, timestamp and timestamp type.
- Offset and partition are what make the read replayable, and they belong in your bronze table.

> ON THE EXAM
>
> A question showing a Kafka read whose value column is treated as a string without a cast is describing code that does not work.

The table-valued functions and their limits

- **read_kafka** works in **both batch and streaming** queries.
- **read_pulsar** is **streaming only**, and in Public Preview.
- **read_pubsub** is streaming only, and works **only inside Lakeflow pipelines**.
- Streaming in SQL at all requires a pipeline, or a streaming table in Databricks SQL.

> ON THE EXAM
>
> These asymmetries are exactly the kind of detail a professional exam tests. Kafka is the flexible one; the others are narrower.

The append-only assumption

- A streaming query over a Delta table picks up new records **when a version is committed**.
- **By default, streaming queries expect the source to contain only appended records.**
- An update or delete in the source will fail the stream, not silently pass through.
- For sources with updates and deletes, Databricks recommends pipelines with **AUTO CDC**.

> ON THE EXAM
>
> This is the same idea as module 5's process-once semantics, seen from the source side rather than the target side.

Lakeflow Connect

- Governed connectors that **extend Auto Loader** to sources beyond object storage.
- The **SFTP** connector supports the same eight formats, with incremental ingestion and exactly-once.
- Credentials and access are governed by **Unity Catalog** rather than living in a notebook.
- Schema inference, evolution and data rescue come along too. It behaves like Auto Loader.

> ON THE EXAM
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> When a scenario names a legacy delivery mechanism such as an SFTP drop, this is the modern answer rather than a scripted download.

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What to carry into the exam

- Parquet, ORC and Avro **carry a schema**. JSON, CSV and XML do not, and that is where inference cost and evolution risk live.
- Kafka returns **bytes**. Cast the key and value explicitly.
- Auto Loader tracks files in its **checkpoint**; COPY INTO tracks them in the **target table**.
- Databricks now points SQL users at **streaming tables** rather than COPY INTO.
- Delta streaming sources are **append-only by default**. Updates need AUTO CDC.

> ON THE EXAM
>
> Set the real format rather than reading as text first. That one habit preserves inference, evolution and rescued data.

> A team reads a Kafka topic carrying JSON payloads and writes to a bronze streaming table. The code selects the value column directly and applies a JSON parsing function to it.
>
> The job fails with a type error, and the developer insists the topic definitely contains JSON text.

What is wrong?

A. The topic must be configured with a schema registry before Databricks can read it.

B. The value column is returned as binary and must be cast to string before parsing.

C. JSON payloads require the Avro deserialiser, not the JSON parser.

D. Streaming reads from Kafka must specify a schema location, as with Auto Loader.

◆ PAUSE AND DECIDE

What is wrong?

A The topic must be configured with a schema registry before Databricks can read it.

A registry helps with Avro. It is not required to read a topic at all.

B The value column is returned as binary and must be cast to string before parsing.

The Kafka reader always returns key and value as byte arrays, whatever was written.

C JSON payloads require the Avro deserialiser, not the JSON parser.

Different format entirely. from_avro is for Avro payloads.

D Streaming reads from Kafka must specify a schema location, as with Auto Loader.

Schema location is an Auto Loader option for files, not a Kafka one.

ON THE EXAM

The developer is right about the topic and wrong about the reader. What was written and what is returned are different questions.

A pipeline ingests daily CSV extracts. Schema inference samples the files and infers a product code column as an integer. Three months in, one supplier begins sending codes that contain letters.

Records from that supplier stop appearing in the silver table. Nothing errors, and the pipeline reports success.

What is the most likely explanation?

A The malformed values are being captured in the rescued data column rather than the typed column.

B CSV files cannot contain mixed types, so the whole file is being skipped.

C Schema evolution added a new column, and downstream queries are reading the old one.

D The supplier's files are being deduplicated because the codes now collide.

◆ PAUSE AND DECIDE

What is the most likely explanation?

A The malformed values are being captured in the rescued data column rather than the typed column.

Values that do not fit the inferred type are rescued rather than failing, which is why nothing errored.

B CSV files cannot contain mixed types, so the whole file is being skipped.

Files are not skipped for type mismatches, and nothing reported a skip.

C Schema evolution added a new column, and downstream queries are reading the old one.

Closer to plausible, but the column already existed. Its type is the problem.

D The supplier's files are being deduplicated because the codes now collide.

Nothing here deduplicates on a product code.

ON THE EXAM

Silence again. Rescued data is designed to avoid losing records, and it succeeds so quietly that nobody notices until a report is short.

A team must ingest a one-off historical load of 900,000 Parquet files from a decommissioned system, then never read that location again. They are comfortable in SQL and want the simplest thing that works.

A colleague argues they should build an Auto Loader stream because it is the recommended tool.

What is the better assessment?

A The colleague is right; Auto Loader is always preferred for file ingestion.

B COPY INTO suits a bounded one-off load, is idempotent on rerun, and needs no checkpoint to maintain.

C Neither works at this scale; the files must be loaded with a Spark batch read and an insert.

D Auto Loader is required because COPY INTO does not support Parquet.

◆ PAUSE AND DECIDE

What is the better assessment?

A The colleague is right; Auto Loader is always preferred for file ingestion.

The recommendation is for incremental ingestion of arriving files. This load is bounded and finished.

B COPY INTO suits a bounded one-off load, is idempotent on rerun, and needs no checkpoint to maintain.

Idempotence covers the retry risk, and there is no ongoing stream to operate afterwards.

C Neither works at this scale; the files must be loaded with a Spark batch read and an insert.

Both handle far more than this. Auto Loader alone scales to billions of files.

D Auto Loader is required because COPY INTO does not support Parquet.

Both support the same eight formats, Parquet included.

ON THE EXAM

Recommendations have scope. Auto Loader is recommended for incremental ingestion; this load has a defined end.

A silver streaming job reads from a bronze Delta table. An operations team began running corrective UPDATE statements against bronze to fix a data quality issue.

The silver stream now fails on every run with an error about the source table.

What is happening, and what is the recommended remedy?

A Streaming reads expect append-only sources. Use a pipeline with AUTO CDC to handle updates.

B The corrective updates broke the checkpoint. Delete it and restart the stream.

C Streaming reads require change data feed to be enabled on every Delta source.

D The stream should be switched to a batch read, since Delta cannot stream after an update.

◆ PAUSE AND DECIDE

What is happening, and what is the recommended remedy?

A Streaming reads expect append-only sources. Use a pipeline with AUTO CDC to handle updates.

The default assumption is append-only, and AUTO CDC is the documented answer for sources with updates and deletes.

B The corrective updates broke the checkpoint. Delete it and restart the stream.

Deleting the checkpoint reprocesses everything, and the next update breaks it again.

C Streaming reads require change data feed to be enabled on every Delta source.

Not required for ordinary append-only streaming reads.

D The stream should be switched to a batch read, since Delta cannot stream after an update.

Delta streams fine. The constraint is about updates in the source, not about Delta.

ON THE EXAM

The failure is protective. Passing an update through as if it were an append would silently corrupt the target.

A regulated insurer receives daily claim files by SFTP from twelve brokers, in a mix of CSV and XML. A scheduled virtual machine currently downloads them with a shell script and copies them to object storage, using a shared password stored in the script.

Auditors have raised the credential handling, and the team also wants schema evolution and rescued data.

What is the strongest recommendation?

A. Keep the script but move the password into a Databricks secret scope.

B. Use the Lakeflow Connect SFTP connector, which extends Auto Loader with Unity Catalog governance.

C. Have the brokers switch to writing directly into cloud storage, then use Auto Loader.

D. Use COPY INTO against the SFTP path directly, since it supports both formats.

◆ PAUSE AND DECIDE

What is the strongest recommendation?

A. Keep the script but move the password into a Databricks secret scope.
Fixes the credential and leaves the script, with no schema evolution or rescued data.

B. Use the Lakeflow Connect SFTP connector, which extends Auto Loader with Unity Catalog governance.
Governed credentials, both formats supported, and Auto Loader's evolution and rescue come along.

C. Have the brokers switch to writing directly into cloud storage, then use Auto Loader.
Technically ideal and depends on twelve external parties changing their systems.

D. Use COPY INTO against the SFTP path directly, since it supports both formats.
COPY INTO reads cloud storage and volumes, not SFTP servers.

ON THE EXAM

Three requirements: governed credentials, both formats, and Auto Loader's schema handling. Only one option delivers all three.

A team must ingest scanned claim documents as PDFs so a downstream process can extract text. They also need to record which file each extraction came from and when it landed.

A junior engineer proposes reading them with the text format and splitting on newlines.

What should they use instead?

A. The text format is correct; PDFs are text files with binary headers.

B. The binary file format, which returns content along with path, length and modification time.

C. The XML format, since PDF is a structured document format.

D. COPY INTO with a custom parser registered as a UDF.

◆ PAUSE AND DECIDE

What should they use instead?

A The text format is correct; PDFs are text files with binary headers.

PDFs are binary. Reading them as text corrupts the content.

B The binary file format, which returns content along with path, length and modification time.

Ingests the bytes without parsing, and supplies exactly the provenance columns the team asked for.

C The XML format, since PDF is a structured document format.

Structured is not the same as XML.

D COPY INTO with a custom parser registered as a UDF.

COPY INTO does not run a parser during load, and a UDF here would be the expensive option from module 4.

ON THE EXAM

The second requirement decides it. Binary file is the only option that returns the path and modification time alongside the content.